What does it mean for an integral to be improper?

I. Domain of integral (interval) is infinite

In this case, we are really just looking at limits

II. Discontinuous Function

AKA if the output of the integral is not infinite

If $|\int f(x)\,dx| < \infty$, we say it converges.

If $\int f(x)\,dx = \pm\infty$, we say it diverges

Infinite Intervals: Let a, b, c be real numbers. So long as the limits exist, we define the following improper integrals as

$$\int_a^{\infty} f(x)dx = \lim_{b\to\infty}\int_a^b f(x)dx$$

$$\int_{-\infty}^{b} f(x)dx = \lim_{a\to-\infty}\int_a^b f(x)dx$$

$$\int_{-\infty}^{\infty} f(x)dx = \lim_{b\to\infty}\int_b^c f(x)dx + \lim_{a\to-\infty}\int_a^c f(x)dx.$$

Example: Find

part 1

$$\int_1^{\infty}\frac{1}{x}dx.$$

Then, for $p \neq 1$, find

part 2

$$\int_1^{\infty}\frac{1}{x^p}dx.$$

$$\int_1^{\infty}\frac{1}{x}dx = \lim_{b\to\infty}\int_1^b\frac{1}{x}dx = \lim_{b\to\infty}\ln|x|\Big|_1^b = \lim_{b\to\infty}(\ln|b| - \cancel{\ln|1|}) = \lim_{b\to\infty}\ln|b| = \infty$$

Then we can integrate as a definite integral where the bounds are the real bounds and then evaluate the output as a limit

$\ln|1| = 0$

$p \neq 1$

$$\int_1^{\infty}\frac{1}{x^p}dx = \lim_{b\to\infty}\int_1^b\frac{1}{x^p}dx = \lim_{b\to\infty}\frac{b^{-p+1}}{-p+1}\Bigg|_1^b = \lim_{b\to\infty}\left(\frac{b^{-p+1}}{-p+1} - \frac{1}{-p+1}\right)$$

$$= \lim_{b\to\infty}\left(\frac{b^{-p+1}}{-p+1}\right) - \frac{1}{-p+1}$$

If $p>1$, $(-p+1<0) = \frac{-1}{-p+1} = \frac{1}{p-1}$

If $p<1$, $(-p+1>0) = \infty$

If you get a num ber out of this it converges. if you get infinity it diverges

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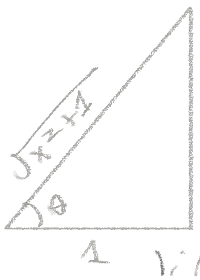
Example: Find

$$\int_{-\infty}^{\infty} \frac{1}{x^2+1}dx.$$

$= \lim_{b\to\infty}\int_0^b \frac{1}{x^2+1}dx + \lim_{c\to-\infty}\int_c^0 \frac{1}{x^2+1}dx$ – we can use trig sub for the integral

How do we get these bands



$\sin\theta = \frac{x}{b}$

$\cos\theta = \frac{1}{b} \rightarrow \cos^2\theta = \frac{1^2}{\sqrt{x^2+1}^2} = \frac{1}{x^2+1}$

$\tan\theta = x$

$\hookrightarrow \frac{dx}{d\theta} = \sec^2 \rightarrow dx = \sec^2\theta\, d\theta = \frac{1}{\cos^2\theta}d\theta$

$\lim_{b\to\infty}\int_0^b \cos^2\theta \cdot \frac{1}{\cos^2}d\theta + \lim_{c\to-\infty}\int_c^0 \cos^2\theta\cdot\frac{1}{\cos^2}d\theta$

$\lim_{b\to\infty}\int_0^b \theta\, d\theta + \lim_{c\to-\infty}\int_c^0 \theta\, d\theta = \lim_{b\to\infty}(\arctan(b)-\arctan(0)) + \lim_{c\to-\infty}(\arctan(0)-\arctan(c))$

$\rightarrow \theta = \arctan(x)$

How

$= \frac{\pi}{2}+\frac{\pi}{2} = \pi$ – convergent

Discontinuous Integrands:

which direction is which

1. Suppose f is continuous on $[a, b)$ and discontinuous at b. Then

$$\int_a^b f(x)dx = \lim_{t\to b^-}\int_a^t f(x)dx.$$

2. Suppose f is continuous on $(a, b]$ and discontinuous at a. Then

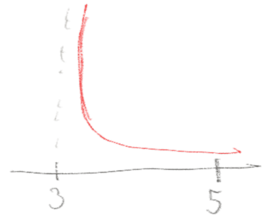
$$\int_a^b f(x)dx = \lim_{t\to a^+}\int_t^b f(x)dx.$$

3. Suppose f is continuous on $[a, c)$ and $(c, b]$ but discontinuous at c. Then

$$\int_a^b f(x)dx = \lim_{t\to c^-}\int_a^t f(x)dx + \lim_{t\to c^+}\int_t^b f(x)dx.$$

Example: Find

$$\int_3^5 \frac{1}{\sqrt{x-3}}\,dx.$$



$= \lim_{t \to 3^+} \int_t^5 \frac{1}{\sqrt{x-3}}\,dx$

$u = \sqrt{x-3}$ $\to$ $du = \frac{1}{2}(t-3)^{-\frac{1}{2}}dx = \frac{1}{2u}\,du$

$\lim_{t \to 3^+} \int_t^5 \frac{1}{u} 2u\,du = \lim_{t\to 3^+} 2u \Big|_t^5$

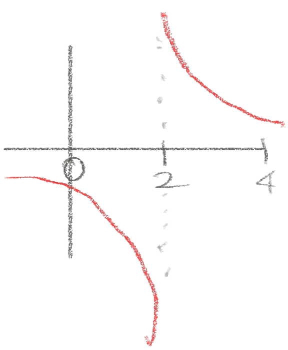
$\lim_{t \to 3^+} 2\sqrt{x-3}\,\Big|_t^5$ $= 2\sqrt{2} - \lim_{t\to 3^+} \underbrace{2\sqrt{t-3}}_{\substack{2\sqrt{\text{almost } 3-3} \\ = 2\sqrt{0} = 0}} = 2\sqrt{2}$

$5-3=2$

Exercise: Find

$$\int_0^4 \frac{1}{x^2-x-2}\,dx. = \int_0^4 \frac{1}{(x-2)(x+1)}$$

So when $x > 2$ then it will be pos
but when $x < 2$ it will be neg $\to$ vertical asymptote



$= \int_0^4 \frac{A}{x-2} + \frac{B}{x+1}\,dx$

$= A\int_0^4 \frac{1}{x-2}\,dx + \underbrace{B\int_0^4 \frac{1}{x+1}\,dx}$

This part is continuous so ignore for now

$A\int_0^4 \frac{1}{x-2}\,dx =$

$\lim_{a \to 2^-} A \ln|x-2|\,\Big|_0^a + \lim_{b \to 2^+} A \ln|x-2|\,\Big|_b^4$

$= A \lim_{a\to 2^-} \ln|a-2| + \ldots$

$= -\infty$ $\to$ The value of the second part won't matter bc you can't cancel out $\pm\infty$

We say that an improper integral is:

- **Convergent** if the integral is finite.
- **Divergent** if the integral is not finite.

Comparison Theorem for Improper Integrals: Suppose f, g are continuous and satisfy

$$f(x) \geq g(x) \geq 0$$

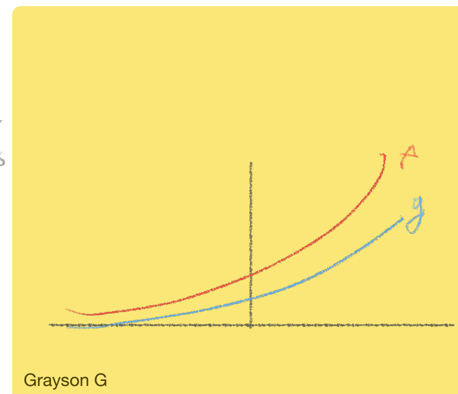
for $x \geq a$. then

1. If $\int_a^\infty f(x)dx$ is convergent, then $\int_a^\infty g(x)dx$ is convergent. If something converges everything smaller converges
2. If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent. If something diverges then everything larger diverges

Example: Show that

$$\int_1^\infty \frac{1+e^{-x}}{x}dx$$

is divergent.



Find something smaller than this that also diverges (something we can easily prove that integral is infinite) → $\frac{1+e^{-x}}{x} \geq g(x) \geq 0$ and $\int_1^\infty g(x)\,dx = \infty$

$e^{-x} \geq 0$ ← Holds for all values x

So smaller function could have 0 instead since e^{-x} is larger than 0 always

Consider $g(x) = \frac{1}{x}$, which satisfies $\frac{1+e^{-x}}{x} \geq \frac{1}{x} \geq 0$ and $\int_1^\infty \frac{1}{x}dx = \infty$

Example: Show that

$$\int_1^\infty \frac{e^{-x}}{x}dx$$

is convergent.

$f(x) \geq \frac{e^{-x}}{x} \geq 0$ and $\int_1^\infty f(x) \neq \infty$

$$0 \leq g(x) = \frac{e^{-x}}{x} = e^{-x}\left(\frac{1}{x}\right) \leq e^{-x} = f(x)$$

If $x \geq 1$ then $\frac{1}{x} \leq 1$

$$\int_1^\infty e^{-x}\,dx = \lim_{b\to\infty}\int_1^b e^{-x}\,dx = \lim_{b\to\infty}\left(-e^{-x}\right)\Big|_1^b$$

$$= \left(\lim_{b\to\infty} -e^{-b}\right) + e^{-1} = 0 + e^{-1} = e^{-1} \leftarrow \neq \infty$$